How Far Can a Reference Be Judged by Its Format? A Controlled Simulation of Format-Only, Title-Only and Full-Field Citation Verification

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Abstract

Fabricated references are hard to spot because they are formatted like real ones. A literature review by the author argued that format inspection alone therefore cannot separate real from fabricated references, and that detection must be grounded in a bibliographic database. This paper tests the computational part of that argument in a small, fully offline, controlled simulation. Using 468 real records from a personal bibliography as a closed-world index, we generate genuine references plus three families of defective ones (invented, chimeric, perturbed-real) and compare four verifiers under a frozen protocol: two format-only methods, a title-only lookup, and a full-field database consistency check. With a format-faithful fabricator, format-only verifiers sit at chance (AUROC 0.504±0.016 for the learned classifier) and gain signal only as the fabricator becomes sloppier (AUROC 0.921 at a 50% malformation rate). A title-only lookup finds invented titles (recall 0.95) but almost never chimeric ones (0.04). The full-field check reaches recall 0.918±0.070 at a 5% false-positive rate, yet its false-positive rate tracks index incompleteness (0.526 at 50% coverage), so at an assumed 2% prevalence its positive predictive value falls from 0.267 to 0.036. Perturbed-real references are hardest (recall 0.756 on average, ranging 0.425–0.906 across seeds), which we attribute, post hoc, to the interaction between the false-positive budget and benign typing noise. All results concern the simulation, not language-model output or human readers; several are consequences of the design, as we state explicitly.

Index Terms— citation verification, fabricated references, hallucination detection, controlled simulation, reproducibility

1 Introduction

Large language models can emit references with plausible authors, titles, venues and years that point to no real work. Walters and Wilder manually checked 636 citations from ChatGPT-written literature reviews and found 55% (GPT-3.5) and 18% (GPT-4) entirely fabricated [1]. A companion review by the author [2] synthesised why such references are believed: their form acts as a peripheral authority cue, and verification effort is rarely applied. That review ended with two computational claims that it could not test itself: (i) because fabricated references are format-faithful, *format inspection alone* cannot discriminate them; and (ii) automated matching can confirm that a source *exists* but not that the reference *as written* is consistent with it, so verification must be grounded in a database.

This paper turns those claims into measurable questions. It is a controlled comparative study (a "Pathway B" study in the course taxonomy), not a new algorithm, and we make *no claim of novelty*: citation-verifier benchmarks already exist, notably HALLMARK [3] and CiteTracer [4]. Our contribution is a small, offline, fully reproducible study that isolates three variables those benchmarks mix together: how much signal *format* carries when fabrications are format-faithful, what a *title-only* lookup adds over a *full-field* check, and how both degrade when the reference database is incomplete.

Research questions. **RQ1:** If fabricated references copy the formatting of genuine ones, how well can format-only signals separate them? **RQ2:** Per fabrication family, what do title-only existence checking and full-field consistency checking each detect and miss? **RQ3:** How do they behave when the index covers only part of the genuine references, and what does that imply for the share of flags that are correct at a realistic prevalence?

Scope and honesty statement. The fabrications here are produced by explicit rules, not by a language model; the index is a closed-world bibliography, not Crossref; and nothing is claimed about human trust. Section 9 lists the threats to validity.

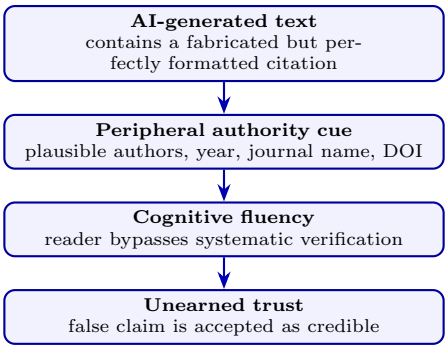


Figure 1: The heuristic-processing loop for fabricated citations, as argued in the author's review [2]. This is the motivating hypothesis; the present study does *not* test human readers.

2 Background and Related Work

2.1 What the review established

Fig. 1 reproduces the review's account of how fabricated references gain unearned trust: correct-looking surface features act as a peripheral cue, are processed fluently, and displace systematic verification. Table 1 restates the review's quantitative evidence with one correction: the review's draft table listed the 35–77% figure of Khurana et al. as an "error rate". It is in fact a language model's hallucination rate on claim-judgement prompts and so measures model behaviour, not the share of references that are fabricated; the table now says what each study measured. The figures in Table 1 were re-checked against the sources' abstracts or pages retrieved during this project.

2.2 Citation-verification benchmarks

HALLMARK [3] offers 2,526 BibTeX entries across 14 hallucination types and three difficulty tiers and evaluates a DOI-lookup baseline, frontier LLMs, tool-augmented agents and a rule-based verifier. Its abstract reports that false-positive rates, not recall, determine whether a verifier's flags are mostly true catches or mostly noise. CiteTracer [4] proposes a 12-code field-level taxonomy, a 2,450-citation synthetic benchmark and a multi-agent detector. Aslett et al. [8] show, for news, that searching online to check a claim can raise belief in it, a caution relevant to how people verify references. Our simulation is far smaller and simpler than these benchmarks; it is complementary in isolating format, title-only versus full-field matching, and index coverage.

3 Hypotheses and Decision Rules

Hypotheses and numeric decision rules were written and hashed before the reported runs (Section 4.4). **H1:** with a format-faithful fabricator the learned format-only classifier has AUROC in [0.45, 0.60]. **H1b:** its AUROC rises by at least 0.15 as the fabricator's malformation rate goes from 0.05 to 0.50. **H2:** the full-field check achieves recall $\geq$0.85 at 5% FPR with a complete index. **H3:** the title-only lookup has recall $\leq$0.20 on chimeric references and on every non-title perturbation. **H4:** with the full-coverage threshold held fixed, the full-field check's FPR is within 0.05 of $(1-c)+0.05c$ at index coverage c. A development smoke run on a separate, never-reported seed was inspected before the thresholds were written; the thresholds are therefore informed expectations, not blind predictions. Two defects were found after the first run and corrected (Section 8).

4 Method

4.1 Corpus and index

The index is `frankenstein.bib` [9], a personal bibliography shipped with TeX Live: 577 bibliographic entries, of which 468 have an author, title, four-digit year and (for articles) a journal, after removing duplicates. These comprise 294 books, 93 articles and 81 chapters or proceedings papers, years 1657–2000. We treat every entry as a real work; this is an assumption (the bibliography is a researcher's library) that we could not verify independently. The file carries no stated licence, so the repository stores only its SHA-256 and a build script, not the file.

4.2 Candidate generation

For each indexed record the generator emits one candidate per class (Fig. 2): **G** (genuine): the record as typed with benign noise; **F1** (invented): authors recombined from name pools, a title from a word-bigram chain rejected if within similarity 0.9 of any indexed title, and venue, publisher and numbers borrowed from a random donor record; **F2** (chimeric): all metadata of record A with the title of record B, echoing the mismatched-field failure described in the review; **F3** (perturbed real): exactly one field of an indexed record altered (year $\pm$1–5, an author surname, one title word, the journal or publisher, or volume and pages), echoing reports of wrong years, volumes and pages [1]. *Benign noise* (initials, dropped optional fields, a title typo, sentence case, dropped subtitle) uses identical rates for all four classes, so presentation style carries no label. *Format malformation* (mangled year, pages, venue, author string, title) is applied per field at rate ρ_g=0.05 to genuine and $\rho_f \in \{0.05, 0.25, 0.50\}$ to defective candidates; ρ_f=ρ_g is the *format-faithful* null condition. The simulation encodes mechanisms from the literature; it is not claimed to match any model's output.

Table 1: Citation-fabrication evidence from the review, restated with what each study actually measured (verified against primary sources).

Study	Systems / sample	Quantity measured	Reported value
Walters & Wilder [1]	ChatGPT-3.5 and -4; 636 citations in 84 papers	Citations entirely fabricated	55% (GPT-3.5); 18% (GPT-4)
Cabezas-Clavijo & Sidorenko-Bautista [5]	Eight consumer chatbots	Reference fully correct; flawed or fabricated	26.5% fully correct; ≈40% flawed or fabricated
Khurana et al. [6]	Seven LLMs; 220,564 prompts; four domains	*Model* hallucination rate on claim-judgement prompts (not a rate of fabricated references)	Up to 35–77% (general knowledge) when a fabricated citation accompanies a true claim; +3 to >22 pp over a no-citation baseline
Liu et al. [7]	U.S. court filings	Filings containing AI-fabricated citations (a count)	Over 1,000; growing year over year

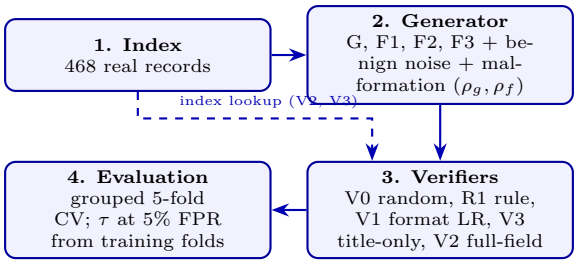


Figure 2: Experimental pipeline. Variants of one source record always share a cross-validation fold (no leakage).

4.3 Verifiers

V0 is a random scorer. **R1** is a naive rule: accept a reference iff it looks well-formed (valid year, no bare "et al.", no all-caps title, valid pages, and so on). **V1** is an L2 logistic regression on 29 structural features (counts, validity flags, presence flags, type); it never sees the index or lexical title content. **V3** retrieves the top five index titles by character TF-IDF and scores the best title similarity (an *existence* check). **V2** scores the same candidates by a fixed weighted consistency over title (0.40), authors (0.30), year (0.10), venue or publisher (0.10) and volume or pages (0.10), renormalised over fields present. V2 and V3 are simplified stand-ins for database cross-checking; they are *not* reproductions of any published tool.

4.4 Protocol and metrics

Five seeds (0–4) each regenerate all candidates and folds. Folds are grouped by source record (variants of one record never straddle train and test). Every operating threshold is the 5th (or 95th) percentile of genuine *training* scores, so each verifier is held to a 5% target FPR chosen without looking at test data. The *primary metric* is out-of-fold recall of defective references at that operating point; secondary metrics are mean-over-folds AUROC, per-family recall and PPV at an assumed 2% prevalence, computed as $\mathrm{PPV} = \frac{r\,\pi}{r\,\pi + f\,(1-\pi)}$ with recall r, FPR f and prevalence π=0.02. In the coverage sweep a random fraction c of index records is removed while thresholds stay fixed at their full-coverage values. We report mean ± SD over five seeds and claim no significance tests. The protocol file and its hash were frozen before the reported runs.

4.5 Reproducibility

Python 3.12.3, scikit-learn 1.8.0 [10], NumPy 2.4.4, pandas 3.0.2, one CPU core; all five seeds and three fidelity levels run in about two minutes (`python scripts/run_experiments.py`). Figures and tables are generated from the saved CSVs; no result is typed by hand.

5 Baseline Validation

Before any comparison we checked behaviour, not just execution, on a non-reported seed: exact copies of all 468 records score 1.0 under V2 and V3 and retrieve the correct record every time; candidate generation is bit-for-bit deterministic; folds share no group; and V1 trained on shuffled labels gives AUROC 0.509 (range 0.473–0.527 over five shuffles), as it must. Altering one field of a clean probe lowers the V2 score by 0.107 (year, weight 0.10), 0.313 (first author, weight 0.30) and 0.068 (venue or publisher, weight 0.10); the last is below its weight because a different venue string still earns partial fuzzy credit, which we flag as a limitation of V2. Twelve unit tests pass; three are regression tests added after defects D1 and D2 (Section 8), which the original nine tests had not caught.

6 Results

6.1 Main comparison (RQ1, RQ2)

Table 2 gives the primary results for the format-faithful condition with a complete index. V2 attains recall 0.918±0.070 at FPR 0.052±0.004 (AUROC 0.985±0.002). V3 attains recall 0.406±0.012; the two format-only methods are at chance (V1 AUROC 0.504±0.016, R1 AUROC 0.497±0.006), and R1 achieves

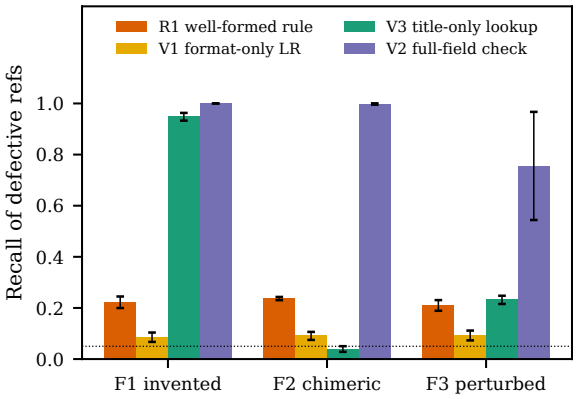


Figure 3: Recall of defective references at the 5%-FPR operating point, by fabrication family (mean ± SD, 5 seeds). Dotted line: chance (0.05).

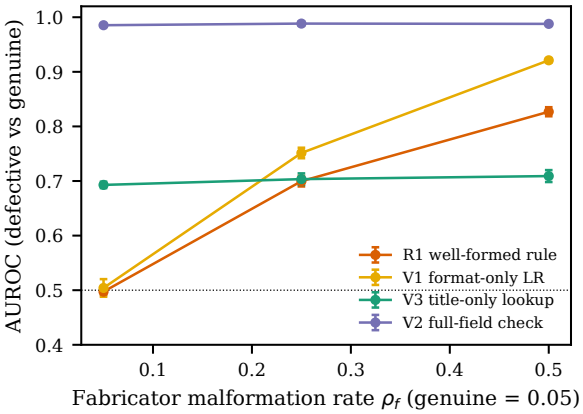


Figure 4: AUROC (defective vs genuine) as the fabricator's format-malformation rate ρ_f grows; genuine references stay at 0.05.

its recall only by flagging 0.228±0.008 of genuine references. V1's realised FPR (0.085±0.013) exceeds its 5% target, so its thresholds do not transfer perfectly from training to test.

Fig. 3 and Table 3 show where each verifier succeeds. Title-only lookup is excellent on invented references (recall 0.95) but nearly blind to chimeric ones (0.04), because a chimeric reference carries a real title; it also misses perturbations of every field except the title. Full-field checking catches invented and chimeric references essentially always (1.00, 1.00) but misses many perturbed-real ones, mainly one-word title changes (recall 0.611) and venue or publisher swaps (0.579).

6.2 Fabricator format fidelity (RQ1, robustness)

Fig. 4 varies the fabricator's malformation rate. Format-only methods gain signal only when the fabricator is sloppier than genuine authors: V1's AUROC rises from 0.504±0.016 to 0.751 and 0.921 as ρ_f goes 0.05, 0.25, 0.50, and the rule R1's to 0.827 at 0.50. V2 is essentially unaffected (0.985±0.002 to 0.988) and V3 is flat (about 0.70).

6.3 Index coverage (RQ3)

Fig. 5 shows that with thresholds fixed, V2's false-positive rate rises almost exactly as $(1-c)+0.05c$: 0.146 at c=0.9, 0.526 at c=0.5 and 0.768 at c=0.25. Recall *rises* (to 0.980 at c=0.25) only because references whose true record was removed can no longer be matched and are flagged, so the gain is not detection skill. At an assumed 2% prevalence the PPV of V2 falls from 0.267 (complete index) to 0.115 (90%), 0.036 (50%) and 0.025 (25%): nearly every flag is then a false alarm.

6.4 Error analysis: why perturbed-real references are hard

V2's F3 recall varies from 0.425 to 0.906 across seeds (SD 0.212) while its FPR stays near 0.05. Fig. 6 shows why: perturbed references other than author changes score about 0.89–0.93 (middle half, seed 0), in the same region as the lower tail of genuine references (typo plus dropped subtitle plus malformed fields); author changes score lower (median 0.67) and are almost always caught (recall 0.995). The 5%-FPR threshold falls inside the overlap region and moves between 0.882 and 0.954 across seeds (Fig. 7). In the lowest-recall seed (seed 1) 6.0% of genuine candidates score below 0.90, more than the 5% budget; in the other seeds at most 4.3% do, and F3 recall is at least 0.660. This explanation is *post hoc*, rests on five seeds, and was not tested further. Of the 121 genuine references flagged by V2 over five seeds, 26% had their author list destroyed (a bare "et al."), 38% had a materially altered title, and 31% had only a combination of minor field mismatches.

6.5 Outcome of the frozen decision rules

Table 4 shows that all five pre-specified rules were met. Section 7 explains why that outcome is less informative than it looks.

7 Discussion

What is by design and what is informative. H1, H1b and H4 are close to consequences of the simulation: a format-faithful fabricator is built so that format carries no label; a sloppy fabricator is built to add format signal; and a record deleted from a closed-world index cannot be matched, so FPR must rise with $1-c$. Their confirmation shows the pipeline behaves as intended, not that

Table 2: Main comparison, format-faithful fabricator (ρ_f=0.05), complete index. Mean ± SD over 5 seeds. FPR is on genuine references; R-F1/F2/F3 are recall per defective family at the operating point; PPV assumes 2% prevalence.

Verifier	AUROC	FPR	Recall	R-F1	R-F2	R-F3	PPV@2%
V0 random	0.508±0.015	0.051±0.001	0.050±0.010	0.05±0.01	0.05±0.01	0.05±0.02	0.020
R1 well-formed rule	0.497±0.006	0.228±0.008	0.223±0.005	0.22±0.02	0.24±0.01	0.21±0.02	0.020
V1 format-only LR	0.504±0.016	0.085±0.013	0.089±0.016	0.09±0.02	0.09±0.02	0.09±0.02	0.022
V3 title-only lookup	0.693±0.006	0.038±0.008	0.406±0.012	0.95±0.02	0.04±0.01	0.23±0.02	0.183
V2 full-field check	0.985±0.002	0.052±0.004	0.918±0.070	1.00±0.00	1.00±0.00	0.76±0.21	0.267

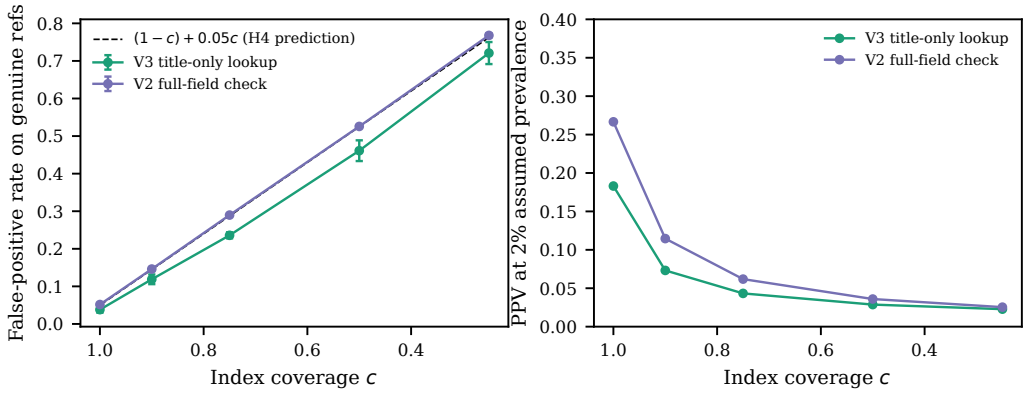


Figure 5: Effect of index incompleteness with full-coverage thresholds held fixed. Left: FPR on genuine references; dashed line is the H4 prediction. Right: PPV at 2% assumed prevalence.

Table 3: Recall on F3 (perturbed-real) by altered field, ρ_f=0.05, mean over 5 seeds.

Verifier	author	year	title	venue/publ.	vol./pages
V1 format-only LR	0.102	0.095	0.114	0.048	0.101
V3 title-only lookup	0.020	0.041	0.846	0.036	0.009
V2 full-field check	0.995	0.833	0.611	0.579	0.706

Table 4: Frozen decision rules and outcomes (mean over 5 seeds).

ID	Frozen decision rule	Observed	Outcome
H1	V1 AUROC at ρ_f=0.05 in [0.45, 0.60]	0.504	supported
H1b	V1 AUROC rise (0.05→0.50) ≥ 0.15	+0.417	supported
H2	V2 recall ≥ 0.85 (FPR 5%, c=1)	0.918	supported
H3	V3 recall ≤ 0.20 on F2 and F3 non-title subtypes	max 0.041	supported
H4	|FPR−[(1−c)+0.05c]| ≤ 0.05 for all c	max dev 0.005	supported

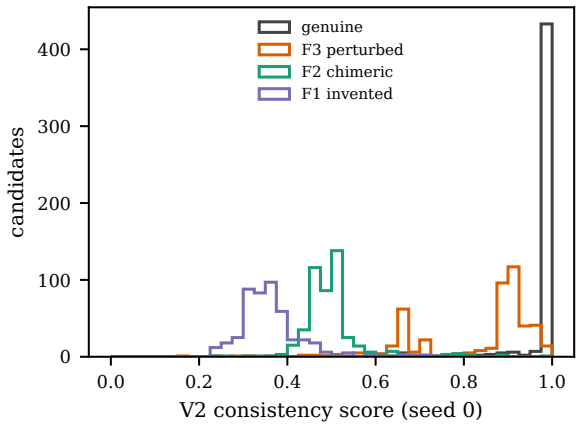


Figure 6: V2 score distributions by candidate class (seed 0). Perturbed-real references overlap the tail of genuine ones.

the world works this way. The informative results are quantitative and non-obvious: (i) the existence-versus-consistency gap is large and measurable (title-only recall 0.95 on invented but 0.04 on chimeric); (ii) the largest residual errors are near-misses (one changed title word or venue), which hide inside the tolerance needed for benign typing noise; (iii) the 5% operating point is unstable when the genuine-score tail is as heavy as the FPR budget; and (iv) PPV collapses with coverage even when AUROC remains high, so a verifier that is excellent in the lab can produce mostly false alarms at deployment.

Relation to the review. The results support the review's computational claims only within the simulation: when fabrications are format-faithful, format-only signals are uninformative, and existence checks are insufficient without field-level consistency. They say nothing about whether *people* are fooled by format; that remains untested here.

Practical implications (untested). Because a missing record is indistinguishable from a fabricated one for a lookup-only tool, reporting flags together with index coverage, and routing "no match" to an UNCERTAIN class instead of "fabricated", appear worth evaluating. We did not evaluate them. The PPV behaviour here

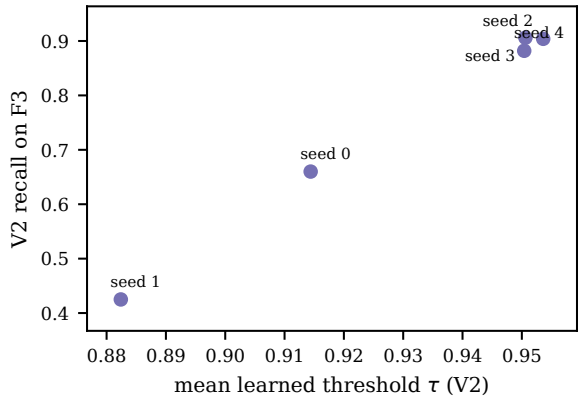


Figure 7: V2 recall on F3 versus the mean learned threshold τ, one point per seed (exploratory).

is consistent in kind with HALLMARK's emphasis on false-positive rates [3], but the numbers are not comparable.

8 Deviations from the Frozen Protocol

Two defects were found *after* the first frozen run, by reading individual false-negative examples rather than aggregate metrics, and all nine original unit tests had passed. **D1 (parser):** an accent-stripping regular expression read the macro `\book{...}` as the accent `\b` plus the letter "o", turning titles such as *Narrative of Molloy* into "Narrative of ookMolloy" in 53 of 468 records. **D2 (label noise):** F3 perturbations were applied before benign noise, so later noise could erase the altered field; 16 of 468 F3 candidates in seed 0 (3.4%) were then identical to the genuine record yet labelled defective. Both were fixed, regression tests were added, all five seeds were re-run with unchanged decision rules, and the first-run results were archived. Table 5 shows the effect: the largest change in any listed metric is 0.041, V2's F3 recall fell from 0.796 to 0.756, and no decision-rule outcome changed. Because the defects were found by inspecting results, the re-run is not blind to the first run.

9 Threats to Validity and Limitations

Simulated fabrications: rules, not an LLM; real fabrications may have artefacts our generator lacks or lack ones it has. *Corpus:* a single personal bibliography, mostly humanities books (1657–2000), no DOIs, 93 articles; the review concerns modern journal references with DOIs, so external validity is limited. *Ground truth:* "real" means "in the index"; entries were not independently verified. *Closed world:* genuine candidates always have an indexed record at full coverage; real databases have coverage gaps and noisy matches. *Parsing:* verifiers receive structured fields; parsing errors from raw reference strings are not modelled. *Hand-set weights and perturbations:* V2's weights were fixed, not tuned, and F3 subtypes are equally weighted; V2 gives partial credit to different venue strings. *Calibration:* V1's realised FPR (0.085 ± 0.013) exceeds the 5% target. *Statistics:* five seeds, SD reported, no significance tests, and the τ explanation is post hoc. *Protocol:* the smoke seed was inspected before decision rules were written, and defects D1 and D2 were found and fixed after the first run (Section 8). *Baselines:* V2 and V3 are not reproductions of published tools and their numbers should not be read as tool performance.

Table 5: Effect of correcting defects D1 and D2 (ρ_f=0.05, complete index, mean over 5 seeds).

Verifier	Metric	Pre-fix	Post-fix	Δ
R1	AUROC	0.502	0.497	-0.005
R1	Recall	0.224	0.223	-0.001
R1	Recall F3	0.221	0.210	-0.012
V1	AUROC	0.498	0.504	+0.006
V1	Recall	0.084	0.089	+0.005
V1	Recall F3	0.085	0.092	+0.007
V3	AUROC	0.700	0.693	-0.007
V3	Recall	0.410	0.406	-0.004
V3	Recall F3	0.240	0.232	-0.008
V2	AUROC	0.981	0.985	+0.005
V2	Recall	0.931	0.918	-0.013
V2	Recall F3	0.796	0.756	-0.041

10 Conclusion

In a controlled, offline simulation, format-only verification is at chance when fabrications are format-faithful and improves only as fabricators get sloppier; title-only existence checking cannot see chimeric or single-field defects; and full-field database consistency detects most defects (recall 0.918 ± 0.070 at 5% FPR) but is hostage to index completeness, with PPV at 2% prevalence falling from 0.267 to 0.036 at half coverage. Natural next steps are (1) real LLM-generated fabrications, (2) a modern DOI-bearing corpus checked against a live bibliographic API, and (3) a three-way verdict (verified, contradicted, unverifiable) evaluated against the coverage effect. Code, protocol, logs and the AI-assistance log are in the accompanying repository.

AI-Use Disclosure

This project was developed with AI assistance (Claude, Anthropic) for planning, code generation, debugging, analysis scripts, figure generation and drafting. All reported numbers are produced by the released code;

sources were checked against primary pages during the project, and the full prompt and verification record is in `ai_assistance_log.md`. The author is responsible for the content.

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